# Project 2 – Image Classification with Convolutional Neural Networks

Due: 7/1/2026

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In this project, we'll take a look at the literature on convolutional neural networks (CNNs), specifically the famous AlexNet and ResNet architectures; and see their implementations and the data pipelines employed to train them. The goal is to get familiar with reading research papers and understanding the implementation details.

## 1. Convolutional Neural Networks

First, we need to understand what a Convolutional Neural Network is. A CNN is a type of neural network that is particularly effective for image-related tasks, e.g. classification, generation, and segmentation. It consists of multiple layers, including convolutional layers, pooling layers, and fully connected layers, which work together to extract features from input images and make predictions. They are more suitable for image data than feedforward neural networks because (1) they can capture spatial hierarchies in images (coarse features such as faces and fine features such as eyes), (2) they require fewer parameters due to weight sharing, and (3) they are more robust to translational variations in the input data (for example, if a car moves from the left side of an image to the right, the network's prediction should not change).

**Task 1.** Explore and understand CNNs in more detail online. I've found the following blog post as a useful first step: Convolutional Neural Networks: A Comprehensive Guide.

## 2. Literature Review

The first CNN was introduced in 1998 by Yann LeCun, and it was called LeNet-5. It was designed for handwritten digit recognition and was trained on the MNIST dataset. Since then, CNNs have evolved significantly, with the introduction of deeper architectures such as AlexNet, VGGNet, and ResNet. In this task, we will focus on AlexNet and ResNet.

**Task 2.** Read the following research papers:

1. **AlexNet:** ImageNet Classification with Deep Convolutional Neural Networks
2. **ResNet:** Deep Residual Learning for Image Recognition

## 3. Implementation

The following repository implements many CNN architectures to be trained on the CIFAR10 dataset, including ResNet: kuangliu/pytorch-cifar.

**Task 3.** Explore the codebase. Specifically, focus on:

- `main.py` - implements the `train()` and `test()` functions, data pipeline, etc.
- `models/resnet.py` - the file containing the ResNet model implementation.

Pay extra attention to implementation details, including any mismatches with the ResNet paper (papers and code often clash!), data augmentation, weight initializations, normalization layers, etc.

**Note:** You do not need to run the codebase or train any models.

## 4. Deliverables

- A report that includes your findings from all three tasks.